Considerations about discretization accuracy in the simulation of lightly damped electric circuits †

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Hardware-in-the-loop (HIL) simulation requires a set of Algebraic-Differential equations to be discretize. The trapezoidal rule of integration is typically chosen in most HIL-capable simulation platforms: RTDS [no reference required!] and Plexim's PLECS[1] for example. In the later, the default solver is a so-called '5th order accurate fully-implicit three-stage Runge-Kutta formula (Radau IIA)'. In short, this formulation is more accurate than the trapezoidal method while avoiding some of its numerical instability issues. In this letter, I explain some advantage of this 'Radau IIA' formulation.

What is the order of a discretization method?

The order of a discretization method of a linear system of equations refers to the ability to mimic correctly the Taylor expansion of the (matrix) exponential. For an autonomous linear system of equation of the form $x' = Ax$, the answer is well known in the continuous time: $x(t) = x_0 e^{At}$, with x_0 being the initial condition. This formulation is applicable to either scalar or matrix formulations.

The problem is now to discretize this equation for computation on real-time systems. The problem is also composed by some stability issue of the solution which is recurrent. We end up with a recurrent discrete formulation of the type:

$$x_{t+h} = x_t e^{Ah} \quad (1)$$

There is the Taylor approximation: $e^{Ah} = 1 + Ah + \frac{(Ah)^2}{2!} + \frac{(Ah)^3}{3!} + \cdots$ which is very accurate but terrible for stability. It's nevertheless the benchmark for accuracy.

The commonly used trapezoidal rule of integration transforms (1) is equal to $e^{Ah} = \frac{I+hA/2}{I-hA/2} = 1 + Ah + \frac{(Ah)^2}{2!} + \frac{(Ah)^3}{4} + \cdots$ Notice that the 3rd term of the division does not equal the Taylor approximation: Trapeze is therefore an order 2 approximation to the matrix exponential. Note that the matrix A can be of rank 1 for the simple scalar case.

How stable is a discretization method?

For passive systems, the exponential response of course tends to decay to zero. It seems the world is built around the exponential response because there is energy dissipation everywhere.

In the discrete world however, this is not always the case. Take the trapezoidal rule again. In (1), the scalar or matrix A contains the time constant of the system. A series resistance-inductor system will have for example $e^{At} = e^{-t(\frac{R}{L})}$, which decay to 0 with an R/L time constant. This time constant can be very short as we increase R and at some point, the time constant tends to 0. In the trapezoidal rule, this is what happens:

$$\lim_{A\to\infty} \frac{I + hA/2}{I - hA/2} = -1$$

So instead of decaying the time response to 0, it will oscillate forever : $x_{t+h} = x_t\,(-1)$ (!!!)

Backward Euler is much more stable by comparison. Its discretization formula is $e^{Ah} = \frac{I}{I-hA}$ and the same limit for very small time constant is:

$$\lim_{A\to\infty} \frac{I}{I - hA} = 0$$

And it will not create oscillations.

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The problem with the Backward Euler method is that is not very accurate. Indeed, it can computed that it is an order 1 approximation to the matrix exponential.

$$e^{Ah} = \frac{I}{I - hA} = 1 + Ah + \frac{(Ah)^2}{1} + \frac{(Ah)^3}{1} + \cdots$$

The Radau IIA method can be shown to be of 5th order (by dividing the Padé (2,3) matrix exponential numerator by the denominator like above) and also that its response goes to 0 with very fast time constant. Let's now show these accuracy and stability properties with a simple example.

A simple switched RLC circuit

To describe the above explanations on accuracy and stability, we use the simple switched RLC circuit of Fig.1.

The resonant frequency of this circuit is 2.9 kHz and it is simulated at a time step of 30 micro-seconds. The sampling frequency is therefore 33.3 kHz just a little bit more than 10 times the resonant frequency.

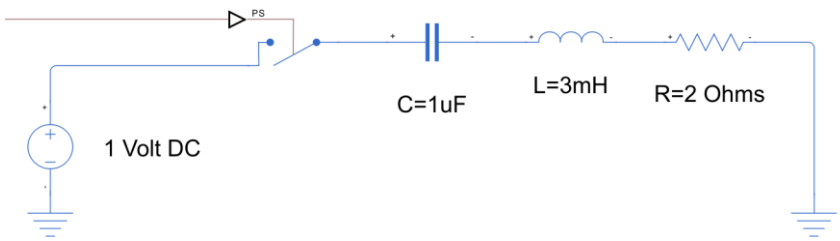


Fig. 1 Simple switched RLC circuit in Simscape Electrical

In the test, we energize the RLC by closing the switch at 10 ms and then re-opening the switch near 20 ms, at the current zero-crossing. One tricky aspect of simulating switched circuits, and breakers in particular, is that you cannot really open them at zero-current crossing. You either have to open the switch after a zero-crossing is detected or you can predict one-step ahead the zero-crossing of the current. But it will (almost) never be zero at the precise time step interval of the simulation.

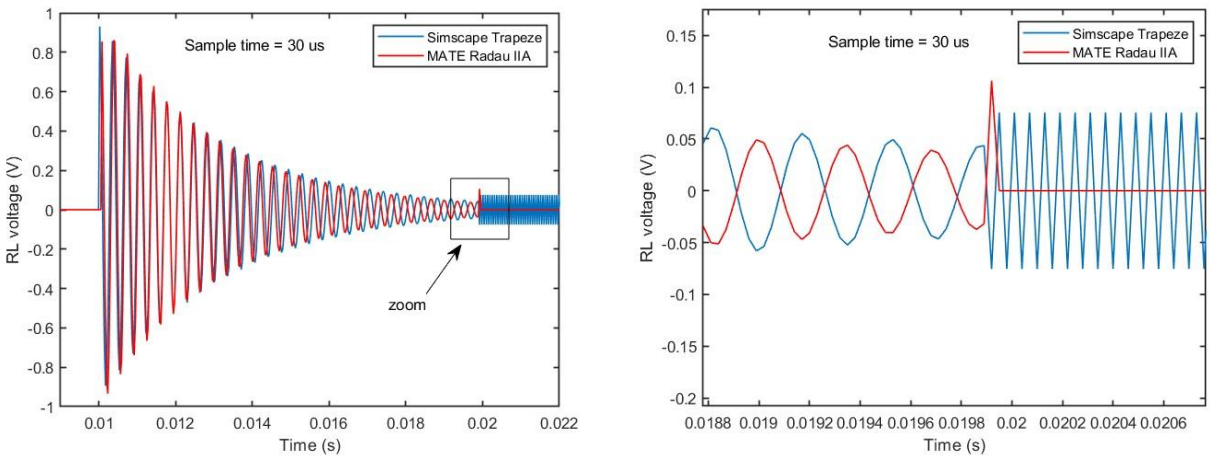


Fig. 2 Step response of the switched RLC circuit: R-L voltage (zoom at right side)

We can observe 2 different things when comparing the Trapeze and Radau IIA simulation results, as shown in Fig.2. First, there is a phase-shift in the resulting voltage, which is really apparent just before the breaker opening. This is the effect of the higher order of the Radau IIA method, which is more evident for frequency component near the sampling frequency of the simulation.

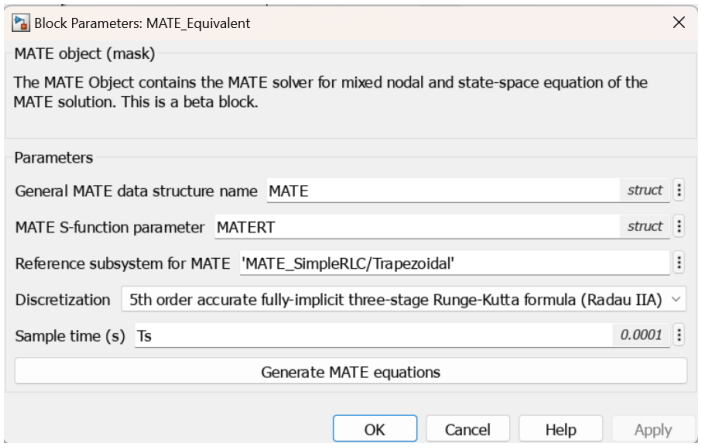
Secondly, there is a voltage chatter at the switch opening for the Trapeze discretization only. This chatter problem is well-known in the EMTP literature. As Fig. 2 shows, the Radau IIA method does not have this problem.

EMTP-type method for solving the chatter issue.

To circumvent this chatter problem some software like EMTP and PSCAD use a dual-discretization method, in which they switch to the Backward Euler method at each time step where a switching action occurs. During non-switching time-step, the discretization is trapezoidal. This solution is excellent for standard simulation software but is commonly implemented in a way that forbids usage in real-time simulators: to keep the Jacobian matrix constant during Backward Euler steps, the sample time is decreased by 2.

How to run the SimScape Electrical example model MATE_SimpleRLC.slx

The model MATE_SimpleRLC.slx compares trapezoidal and Radau IIA (5th order) methods. While the former is simulated with the standard SimScape local solver, the latter requires the RLC circuit to be reformulated into state-space using the MATE Object block. After one dummy simulation run to make sure that Simulink updated correctly all parameters, users must open the MATE_Equivalent block and press the <Generate MATE equations button>.



Conclusion

This letter explained the concepts of accuracy and stability of discretization method for linear system of differential equations. These definitions extend also time-piecewise linear differential equation like switches circuits. It is noticeable that both accuracy and stability issues tend to solve themselves when the simulation time step is decreased. In FPGA simulation with sample time below 1 micro-second typically, these issues are often not critical. But not always…

In some induction stove-top modeling with resonant frequency above 20-25 kHz typically [3], the choice of solver can be important. In can be shown that the Backward Euler method in these simulations is inaccurate even at 1 us [4]. The issue with the methodology in [4] is that the inductance and resistance of the stove-top model are fixed, where in reality they are continuously changing with the operating point. Indeed, it is very challenging to design a variable admittance solver for FPGA simulation. This will be the subject of another letter.

References

[1] PLECS, The simulation platform for power electronic systems, User Manual Version 5.0, Plexim GmbH, Switzerland https://www.plexim.com

[2] J. R. Marti and J. Lin, "Suppression of numerical oscillations in the EMTP power systems," in *IEEE Transactions on Power Systems*, vol. 4, no. 2, pp. 739-747, May 1989,

[3] Qais, Mohammed & Kewat, Seema & Loo, K.H. & Lai, Cheung-Ming. (2023). Early outlier detection in three-phase induction heating systems using clustering algorithms. Ain Shams Engineering Journal. 15. 102467. 10.1016/j.asej.2023.102467.

[4] https://www.mathworks.com/help/hdlcoder/ug/generate-hdl-code-for-simscape-models-using-trapezoidal-rule-solver.html